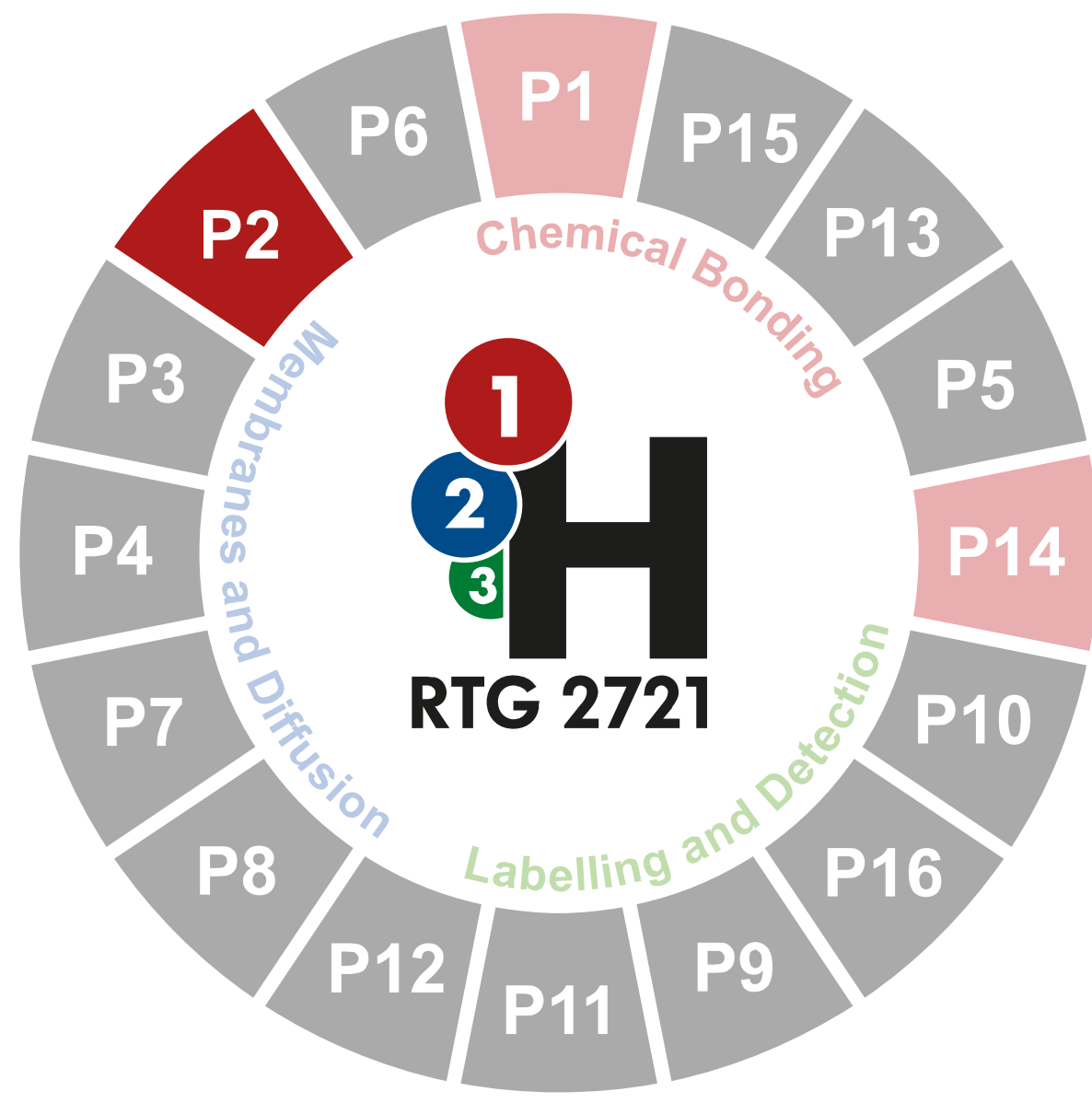




Development of Materials for Light Isotope Selectivity

Elvira Gouatieu Dongmo, Bernd Abel, Thomas Heine*

Motivation and Aim

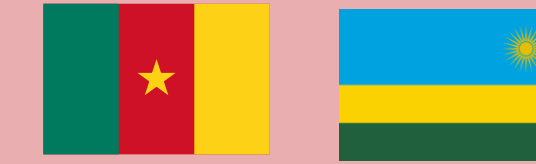
- For decades, the separation of hydrogen isotopologues has been achieved almost exclusively by **cryogenic distillation**, that remains highly energy-demanding. This project employs **cutting-edge computational methods** to investigate the impact of coordination geometries, ligand-sphere environments, pore tuning, on the **separation of hydrogen and helium isotopes**.
- The findings of this study will contribute of fine-tuning open metal sites in porous materials, especially in **metal-organic frameworks (MOFs)** to achieve more efficient **hydrogen and helium isotopes separation**.

Methods and Approach

- Ab initio potential energy surface (PES) computed using **CCSD(T)** and **DLPNO-CCSD(T)** methods.
- Anharmonic vibrational effects: treated via second-order vibrational perturbation theory (**VPT2**)
- Frequencies and zero-point energies (**ZPEs**): obtained by numerically solving the Schrödinger equation using the finite-difference method (**FDM**)
- Software: CFOUR 2.1, Gaussian 16 were employed for all optimizations, energies and frequency calculations and ORCA for single point energy calculations only.

Personal Profile

Origins and Education:



^{1,2,3}H DR: Oct./2021 – Apr./2025

Conference(s):

03 International Symposia on Theoretical Chemistry (58th in 2022, 59th in 2023, and 60th in 2024) with poster presentation.

Research Stay: January 22 – February 04, (2023)

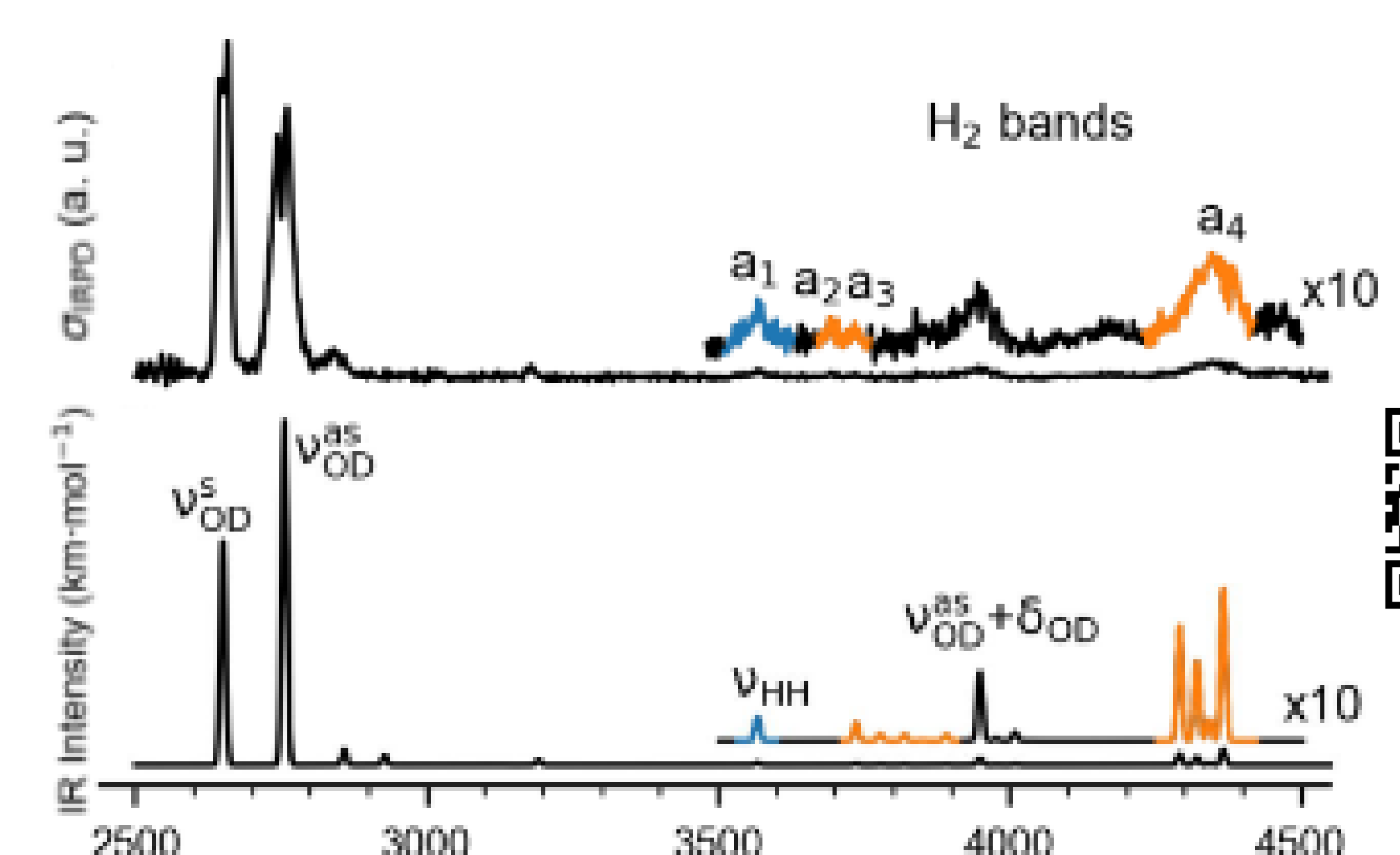
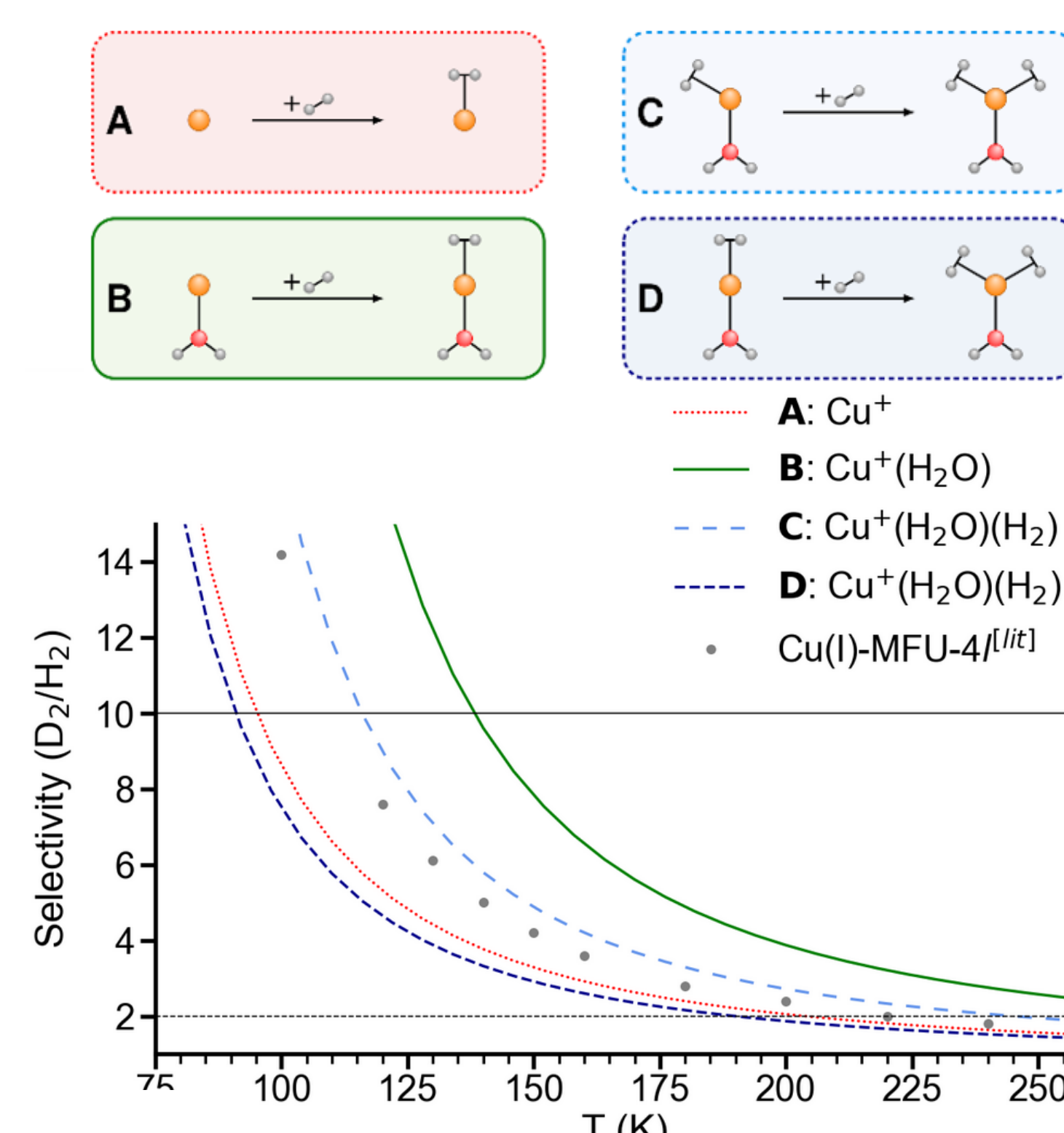
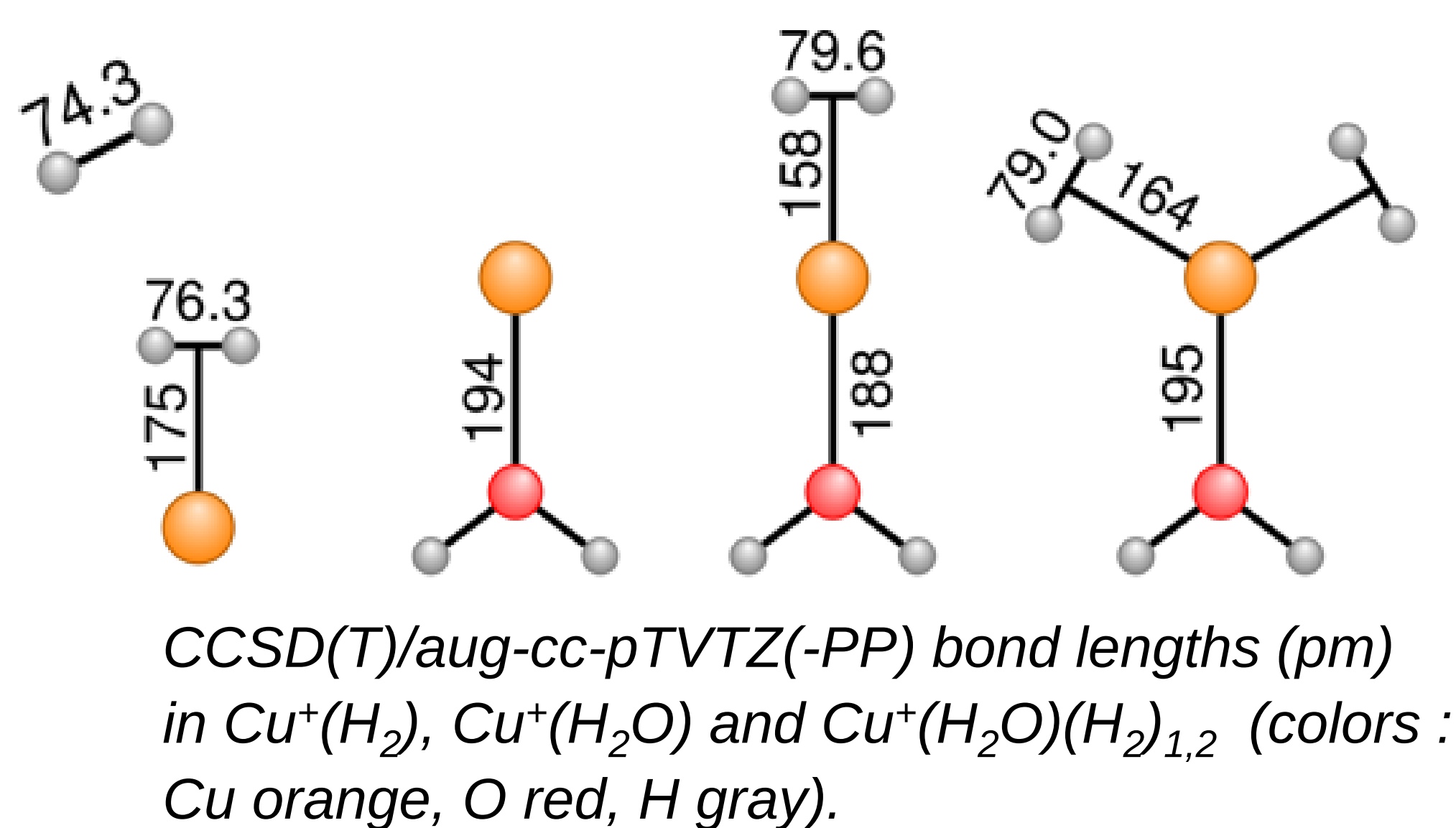
Prof. Dr. A. Daniel Boese (Department of Chemistry, Graz).
Description: establish efficient computational protocol for high-level static vibrational frequencies analysis with Cfour and Molpro software packages.

Major activities:

- LNDW "Chemistry lab of the future-robotic in chemistry";
- Third prize for the three Minute Thesis (3MT), Leipzig (2024)



Direct evidence for ligand-enhanced activity of Cu(I) sites

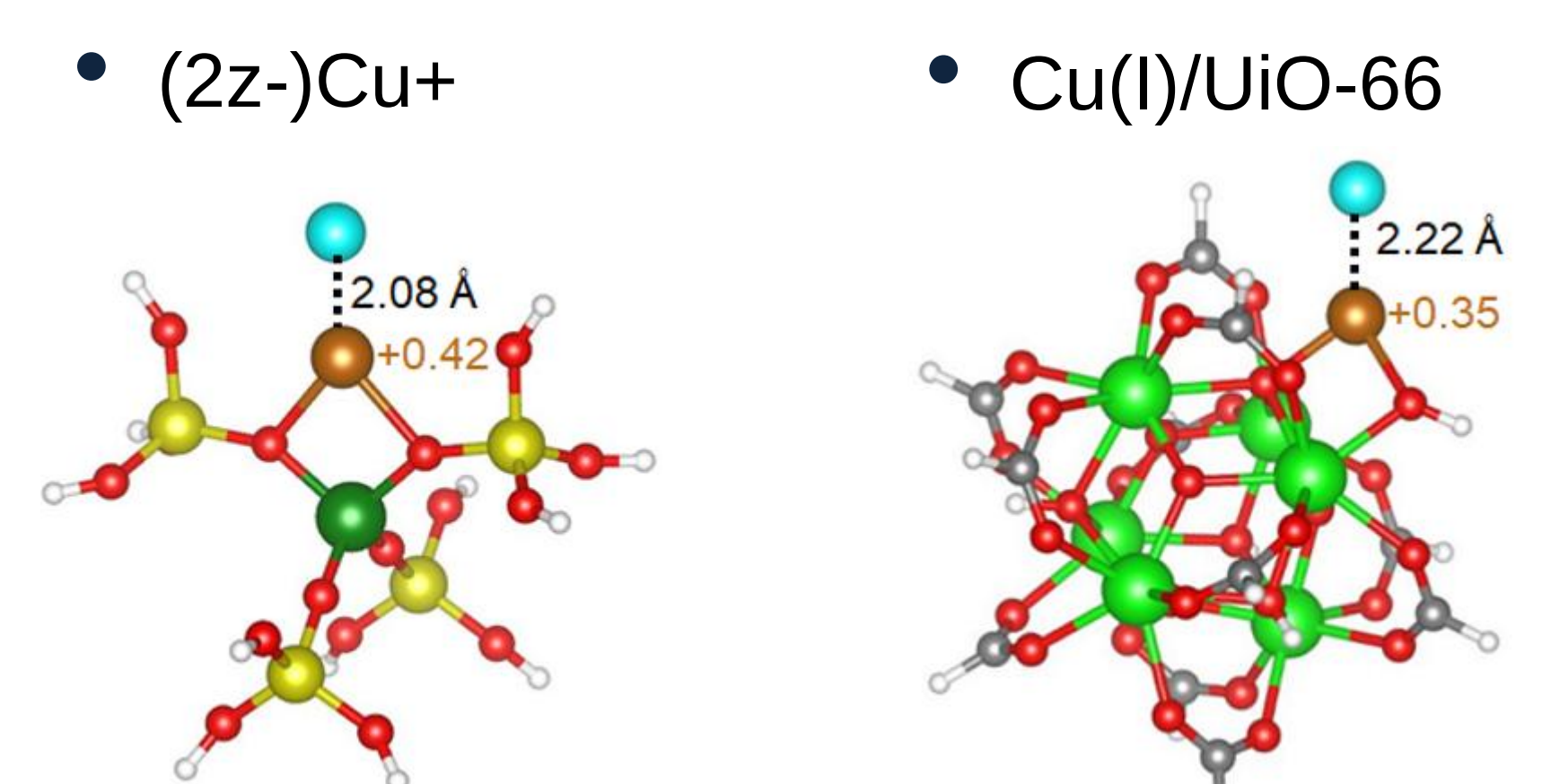
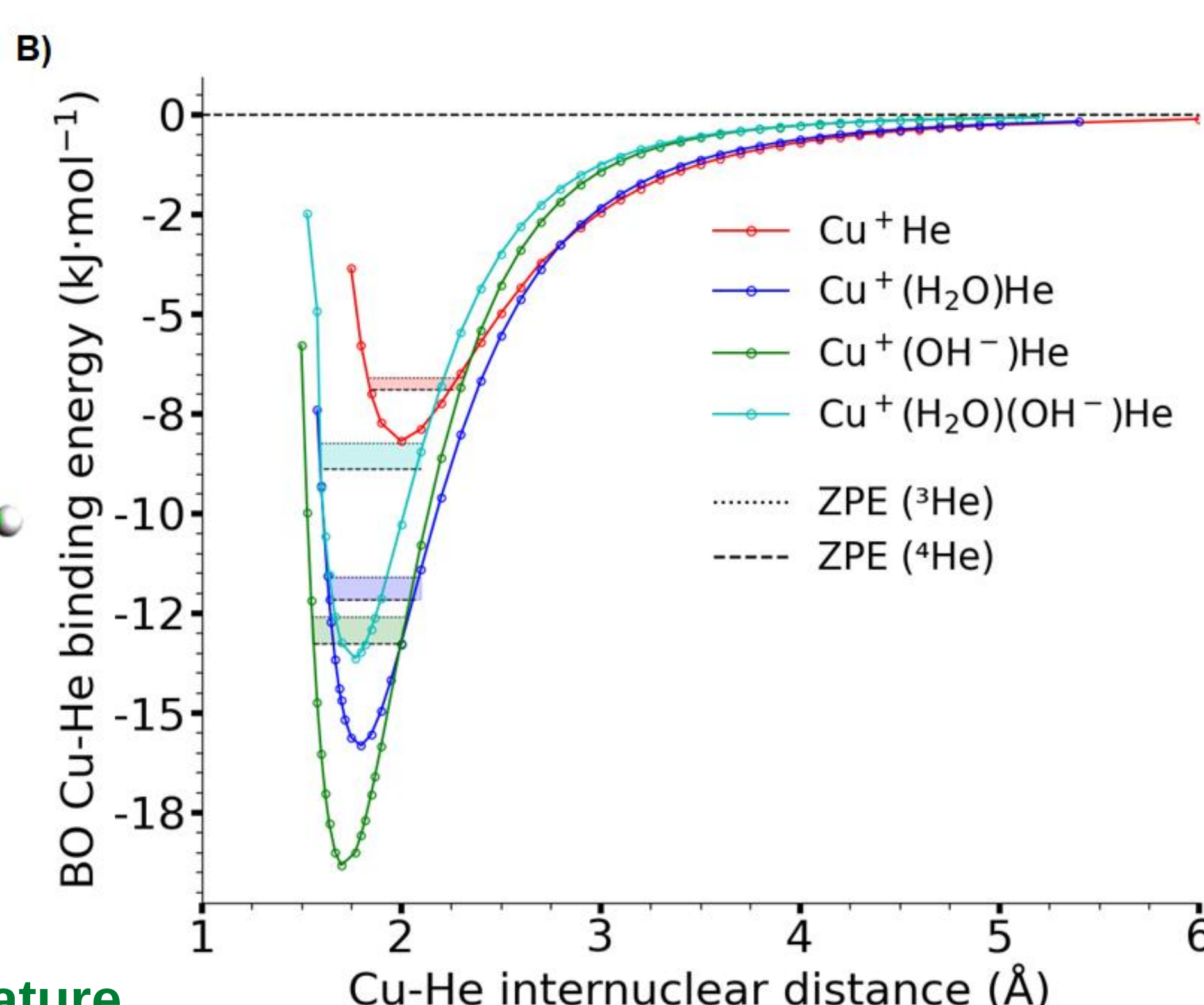
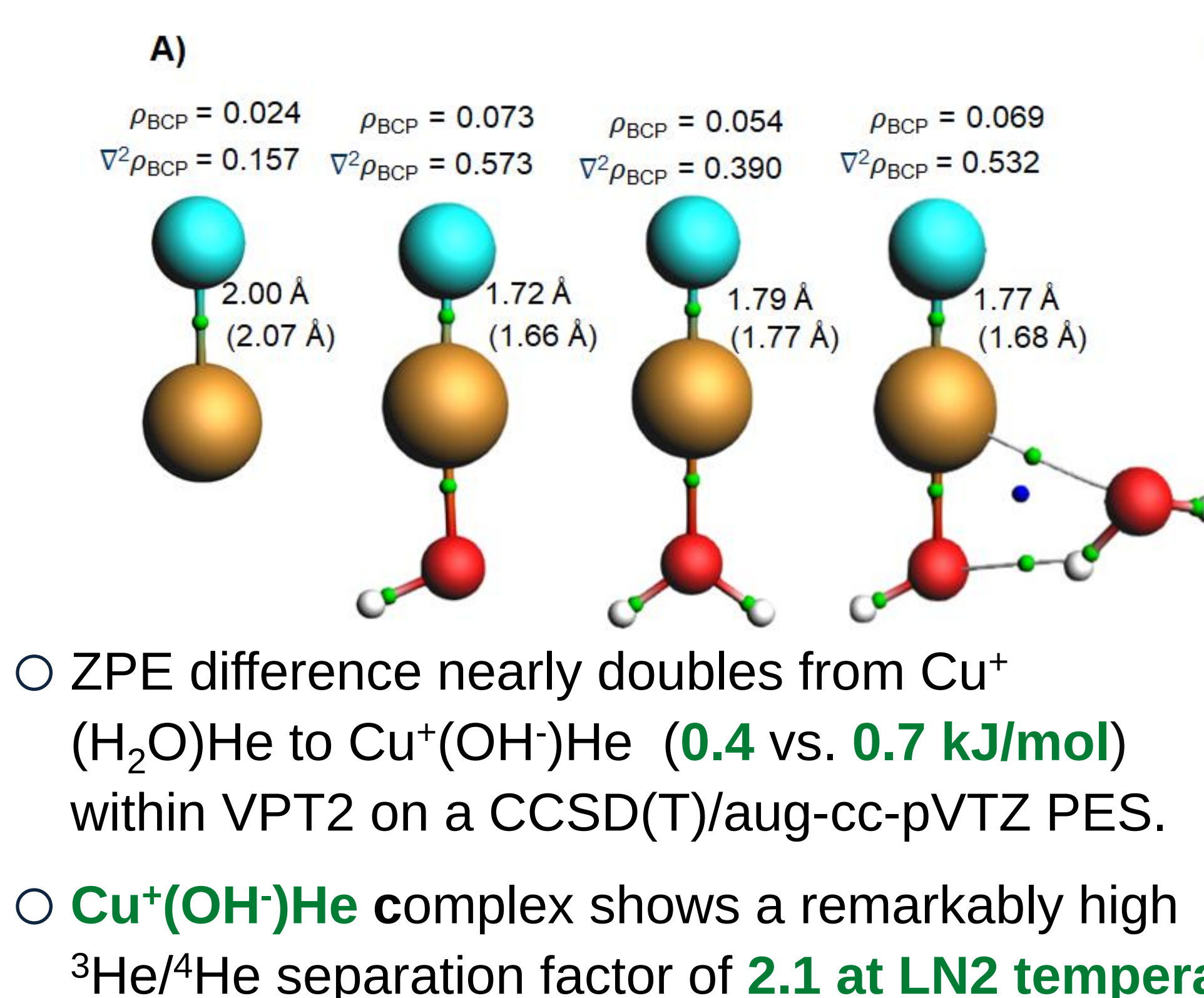


- Bond strengthening between Cu⁺ and H₂ upon addition of a single H₂O ligand.
- Bonding analysis with EDA-NOCVs for Cu⁺-H₂ bond. Collaboration with Florian (P14)

- D₂/H₂ separation factor is **2** at room temperature and rises to **10** at **150 K**.

- IRPD of Cu⁺(D₂O)(H₂)₂ match the calculated IR when taking anharmonicity using VPT2/MP2/def2-TZVPP.
- Combination bands are observed (orange-coloured bands a₂₋₄). Collaboration with Shabnam (P1)

Strong Cu(I)-He interaction at open metal sites enables isotope-selective helium adsorption



- ZPE difference nearly doubles from Cu⁺(H₂O)He to Cu⁺(OH⁻)He (**0.4** vs. **0.7 kJ/mol**) within VPT2 on a CCSD(T)/aug-cc-pVTZ PES.
- **Cu⁺(OH⁻)He** complex shows a remarkably high ³He/⁴He separation factor of **2.1** at **LN2 temperature**

- The difference in ZPEs between ³He and ⁴He plays a key role in determining selectivity.
- Among all the studies material models, Cu(I) anchored at tetrahedral sites in zeolites and in **UiO-66 MOF** emerge as promising candidates, with selectivity close to **3** at **LH2 temperature**.



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